Mk14-GPU

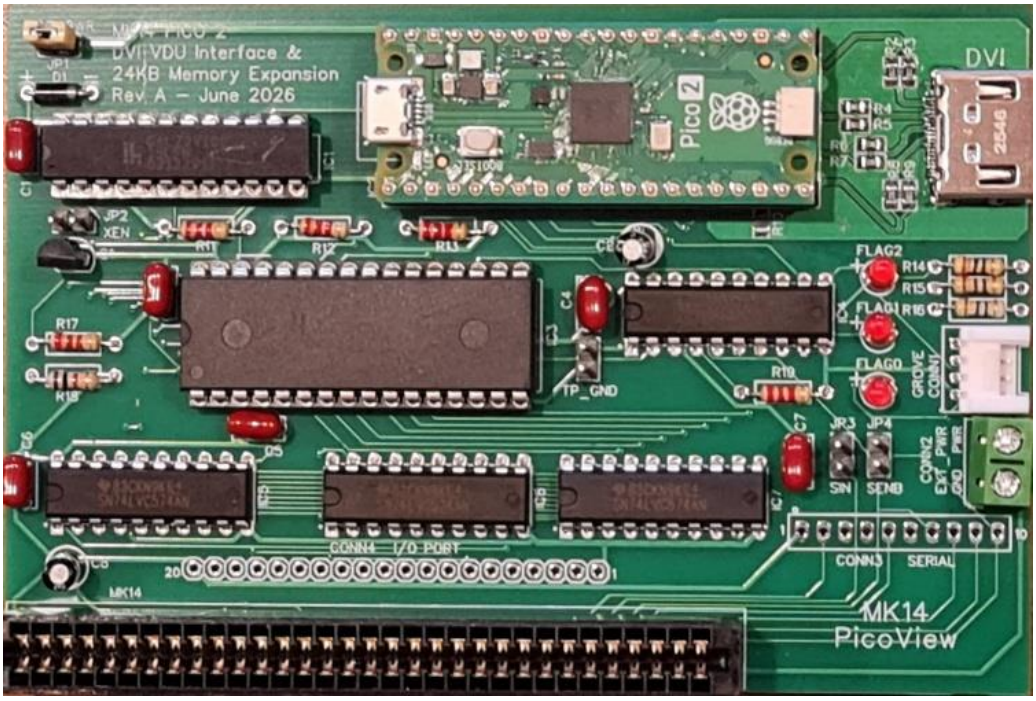
(aka PicoView graphics adapter)

User Manual
for firmware version 0.7.3

1. History

Over the past few years Ian Reynolds and Ian Murray have created VDU boards for the Mk14. The boards were created due to the inability to connect an original MK14 VDU board to a modern TV. The RealView board and the PiView board took different approaches at creating the video output, both boards had their strengths and weaknesses. The Mk14-GPU has been built on the strengths of these two boards, and has the ability to showcase some of the Mk14 programs that have been developed over the past few years.

Design philosophy was to create an Mk14 VDU board that could be used without modification on any Mk14 running SCIOS-2 or SCIOS-3 and with address and data bus available at the edge connector, and plug into any standard Monitor or TV with HDMI input.



2. Mk14-GPU General Description

The Mk14-GPU is a multi-function graphics interface for the Mk14 computer with additional 22K bytes memory. It provides all the text and graphics modes of the original Science of Cambridge (Soc) Mk14 VDU module but adds significant functionality and new operating modes to enhance Mk14 programs and usability.

The Mk14-GPU has the following features:

- Simple interface to the Mk14, with no special hardware modifications required
- HDMI 640 x 480 colour video output
- Supports all standard Mk14 SoC VDU video modes.
  - 16x32 text mode – monochrome or colour
  - 64x64 pixel graphics
  - Mixed 16x16 text (top half of the display) / 64x32 pixel graphics (lower half of the display)
- Additional display modes
  - Solitaire custom colour games font
  - Invaders custom colour games font
  - Connect4 colour display
  - TETRIS colour display
  - 32x48 text mode
  - 128x96 pixel graphics mode
  - Mixed 32x24 text (top half of the display) / 128x48 pixel graphics (lower half of the display)
  - Dual windows
    - Left hand window – standard SoC modes
    - Right hand window – 16x32 text mode
- Colour palette mode shows the hex codes for available colours
- 23.5KBytes SRAM for user programs and data[1]
- Mk14 7-segment LED display is replicated on the Mk14-GPU screen
- Built in FastLoader interface supports Mk14 SCIOS-2 and SCIOS-3 operating system
- 15 high quality programs provided for the fast loader, stored in on-board flash memory
- 3 LED's showing status of SC/MP Flag outputs.

3. Hardware Description

The Mk14-GPU module plugs on the rear connector of the Mk14. It is fully compatible with the Mk14 Issue V[2], Issue VI (Ian K Rolfe) and other Mk14 variants that bring the SC/MP address, data and control signals to the rear connector strip.

The core of the Mk14-GPU is a Raspberry Pi PICO 2 module that decodes the Mk14's buses and generates the video output.

All Mk14 signals are buffered and level translated for compatibility with the PICO2

3.1 Power considerations.

The Mk14-GPU can be powered from the Mk14 directly via the edge connector (Side A, +5V), or Power can be connected to Conn 2 on the GPU board. Current consumption is less than 100mA. Care should be taken to ensure the correct voltage is applied to Conn 2 since this voltage will be sent to the Mk14 via the standard Mk14 edge connector (Top Side Pin1-3 Power +5V, Pins 4-6 Power 0V).

[1] FRAM non-volatile memory is also supported

[2] When the address, data and control signals have been hand wired out to the MK14 rear connector strip

Power to the PICO2 module is derived from either the Mk14 5V supply, via a blocking diode, or from the PICO2 USB interface. When loading firmware into the PICO2 module the Mk14-GPU's PICO_PWR jumper should be removed to allow the PICO2 BOOT button to respond to a power up. For normal operation fit the PICO_PWR jumper.

3.2 Connectors and jumpers

Connector / jumper	Function	Comments
Mk14	Main interface to the Mk14 address, data and control signals	
CONN3: SERIAL	Provides buffered versions of the SC/MP flags and serial interface signals	
CONN4: I/O PORT	The Mk14 parallel I/O signals from the INS8154 are brought out to this header	Allows access to the INS8154 ports and interupt signals when the Mk14-GPU is fitted
CONN1: GROVE	External interface to the FastLoader serial pins	See special consideration in section ???
JP1: PICO_PWR	Supplies 5V to the PICO 2 module.	The default for this jumper is 'fitted' but it can be removed when the PICO 2 USB interface is connected to allow PICO BOOT mode to be entered.
JP3: SIN	Serial Input for the FastLoader interface.	Default for this jumper is 'fitted'. See special considerations in section ???
JP4: SENB	Sense_B input for the FastLoader interface.	Default for this jumper is 'fitted'. See special considerations in section ???
TP_GND	2 pin header for connecting a 'scope probe ground lead, or similar.	Provides a convenient GND access point.
DVI	HDMI Type A connector providing the interface to a standard PC monitor ot TV	640x480 VGA video compatible with HDMI and DVI interfaces.
EXT_PWR		
JP2: XEN	Not used	

4. Mk14-GPU Configuration and Control: 0xF200 - 0xF2FF

A block of memory from 0xF200 to 0xF2FF is reserved for controlling the Mk14-GPU:

0xF200 : Set display mode
0xF201 : Display mode information and SW version. Timeout after 5 seconds
0xF202 : MK14 sets character mode foreground colour using RGB222 values
0xF203 : MK14 sets character mode background colour using RGB222 values
0xF204 : Select program to be fast loaded from Flash to MK14 memory
0xF205 : Display RGB222 colour palette
0xF206 : Display 7 segment display when non-zero. Disable when zero (default is ON)
0xF207 : Display a splash screen graphic for demo purposes. non-zero to enable, 0 to disable
0xF208 : Set fast loading mode: 0x02 = SCIOS-2, 0x03 =SCIOS-3



Set Display Mode

Mode selection is via a byte written to address 0xF200. Modes identified by * can also be selected by writing to address 0x07FF, this allows programs developed for the PiView display adapter to work without modification.

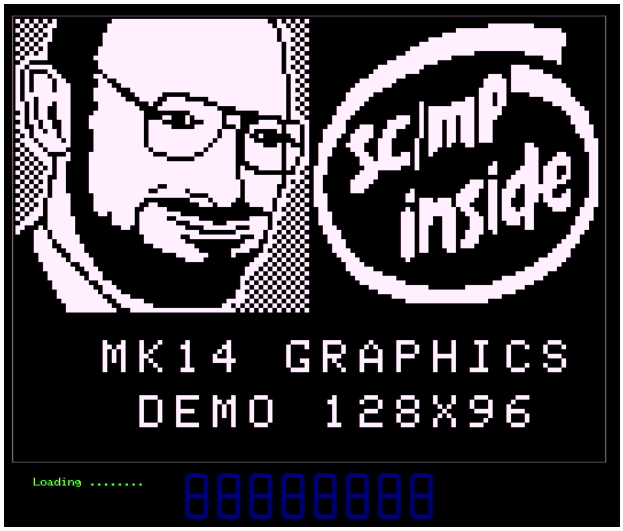
Mode description	Data to write	Comment
***Character Mode 16x32**	0x00	Standard SoC VDU mode
***Graphics Mode 64x64**	0x04	Standard SoC VDU mode
***Mixed Mode 64x64**	0x0F	Standard SoC VDU mode
Dual window Text / Text	0x20	Left and right windows are 16x32 text
Dual window Graphics / Text	0x24	Left = 64x64 graphics, right = 16x32 text
Dual window Mixed / Text	0x2F	Left = mixed text and graphics, right = 16x32 text
Character Mode 32x48:	0x80	Character mode giving an increased text area
Graphics Mode 128x96:	0x84	Hi-res graphics mode
Mixed Mode 128x96:	0x8F	Mixed 32x24 text and 128x48 graphics
***Solitaire colour mode:**	0x0D	PiView game mode, custom games font
***Space invaders colour mode:**	0x0C	PiView game mode, custom games font
***Connect4 colour mode:**	0x0B	PiView game mode
***Tetris colour mode:**	0x10	PiView game mode

At the point of mode change, an information panel will be displayed next to the 7 segment display for 5 seconds.



If a mode change is forced as part of the fast load process (i.e. 0xF200 is included in the hex file) then that line of hex should be separate from other hex data. This allows the Mk14-GPU to clear screen and set up the mode while the next line address bytes are being transferred. Data bytes may be lost in the transfer if this constraint is not observed.

When a mode change is made as part of the execution of an Mk14 program, then a delay DLY 0x40 has to be inserted after any write to 0xF200 to allow the Mk14-GPU to clear the display and set up the mode. If this is not done then writes to screen memory may be lost.





Show Mk14-GPU status window

To force the information box to be displayed for 5 seconds write any value to 0xF201



Text foreground colour

The default colour for text modes is white text on a black background. Location 0xF202 is used to set a specific foreground colour for text modes. Any value in the range 0x00 to 0xFE will change the colour, but refer to the colour palette display for specific colours/hex values.
Writing 0xFF to this location will set the foreground to its default white and background to its default black.



Text background colour

The default colour for text modes is white text on a black background. Location 0xF203 is used to set a specific background colour for text modes. Any value in the range 0x00 to 0xFE will change the colour, but refer to the colour palette display for specific colours/hex values.
Writing 0xFF to this location will set the foreground colour to its default white and the background to its default black.



FastLoader program selection

0xF204 is used to select a program to be fast loaded from the Mk14-GPU flash memory to the Mk14, using the following process:

1. Enter the code for the program you want to load into address 0xF204 (see the table below)
2. The monitor should show "Waiting for 0052" (SCIOS-3) or "Waiting for F700" (SCIOS-2)
3. Hit ABORT and enter appropriate fast loader address.
4. The screen should show "Press Go to load file"
5. Hit GO
6. The screen will display Loading …….

Program selection	**Program selection**
E0 = Fast Loader (for SCIOS 2)	E8 = Text_48x32
E1 = Demo128Gr	E9 = Connect4
E2 = Solitaire	EA = Analogue Clock
E3 = Tetris	EB = Jackpot
E4 = LED Digital Clock	EC = Moonlander Graphics
E5 = Single-Step	ED = Snake
E6 = Invaders	EE = Fallman
E7 = PicoView_LED_Test	EF = Life14
	F0 = Dual window demo



Display the colour palette

Writing any non-zero value to this address will display the Mk14-GPU colour palette and corresponding hex values to set the foreground and background text colours (see F201 and F202 above). Writing 0x00 to this address will clear the screen and revert the display back to the previous display mode.





Enable/disable on-screen seven segment display

Writing 0x00 to this address will blank the on screen seven segment display. Writing any non-zero value will enable the on screen seven segment display.



Enable/disable splash screen graphic

Writing any non-zero value to this address will display the Mk14-GPU splash screen graphic. Writing 0x00 to this address will remove the splash screen graphic.



Set FastLoader SCIOS mode

This address is used to force the fast loader mode. Writing 0x02 sets the mode to SCIOS-2; writing 0x03 sets the mode to SCIOS-3. Writing any other value has no effect.

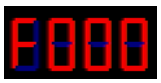
5. Using FastLoader

FastLoader was developed by Ian Murray to reduce the time taken to load programs into the Mk14. The SCIOS-3 was updated to replace the SCIOS-2 tape load and save routines with the FastLoader code at 0x0052. FastLoader uses synchronous data transfer instead of the original Mk14 asynchronous data transfer method, so programs that used to take minutes to load can now be loaded in seconds. FastLoader uses the original SIN and Sense-B interface pins of the original tape interface routines.

If SCIOS-2 is installed in the Mk14, the Fast Loader core routine needs to be downloaded from the Mk14-GPU flash into the GPU RAM. In theory the FastLoader could be installed into any location of the Mk14 memory, but the Mk14-GPU expects it to be loaded into 0xF700. The following procedure should be used to load the FastLoader, when using SCIOS-2.

- After Mk14 reset, enter 0xF7 into 0x0FF9 and 0x00 into 0x0FFA.
- Enter 0xE0 into 0xF204. The screen will display “Waiting for 007C”
- Press Abort, enter 007C and press Go. The Mk14-GPU will display Loading …… and send the FastLoader code to the Mk14 to be stored at 0xF700.
- Press Mk14 Reset when the loading process has finished, i.e. when “Loading …….” is no longer shown on screen.

F700 should be used to load programs into the Mk14, the above procedure will need to be repeated if power to the Mk14 is cycled.



If at any point you want to abort transfer sequence, before a transfer starts, enter any value at 0xF000 to exit the Fastloader process.

If jumpers are inserted into JP3 and JP4 so that the Mk14-GPU FastLoader can be used, it is **essential** that any device connected to the Conn1 Grove connector has tristate outputs, to avoid conflicts with the Mk14-GPU outputs. If that is not the case, disconnect the device from the Grove connector.

6. Memory map

Mk14 fully populated with Mk14-GPU attached

Address **x = 4K page number**	**Function – All pages**[3]
x000-x1FF	512 byte SCIOS ROM
x200-x7FF	GPU RAM - see below
x800-x8FF	128 bytes I/O chip RAM
x900-x9FF	Keyboard & Display
xA00-xAFF	I/O Chip & RAM Shadow
xB00-xBFF	256 bytes RAM (Extended)
xC00-xCFF	I/O Chip & RAM Shadow
xD00-xDFF	Keyboard & Display Shadow
xE00-xEFF	I/O Port & RAM Shadow
xF00-xFFF	256 bytes RAM (Standard)

Mk14-GPU RAM Allocation

Address **x = 4K page number**	**Function: Page 0**	**Function: Pages 1-E**	**Function: Page F**
x200-x2FF	Video frame buffers – see below	User RAM: $600 bytes per page	Mk14-GPU control
x300-x6FF			Reserved for future Mk14-GPU enhancements
x700-x7FF			Reserved for FastLoader code

Mk14-GPU Video frame buffers

Address **(Page 0 only)**	**Standard SoC display Modes**	**Extended Hi-Res display modes**	**Dual window display mode**
0200-03FF	Frame buffer	Frame buffer	Left frame buffer
0400-05FF	User RAM: $400 bytes[4]		Right frame buffer
0600-07FF			User RAM: $200 bytes

[3] All Mk14 ROM, SRAM and I/O devices are mirrored in each 4K page as is normal for the Mk14

[4] This block of RAM is widely used by Mk14 programs